

# Differential CC $1\mu 1\pi^\pm$ cross sections on argon in the NuMI beam

Inclusive measurement and the proton-tagged hadronic-mass / TKI subsample

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$\nu_\mu + \bar{\nu}_\mu$  charged-current single charged pion • FHC, RHC, combined

## Part I — Inclusive $CC1\mu1\pi^\pm Xp$

1. Motivation & signal
2. Beam, exposure, selection
3. Cut-flow & performance
4. Unfolding & response
5. Differential results + generators
6. Integrated & total cross sections
7. Systematics & sideband validation

## Part II — Proton-tagged ( $W$ / TKI)

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5. Background composition

## Part III — Multi-pion ( $2\pi$ , $3\pi$ )

1. Selection tuning & performance
2. Dedicated pion-ID BDT

## Backup

## Signal definition

$\nu_\mu/\bar{\nu}_\mu$  charged-current, exactly one muon and one charged pion, any number of protons, no  $\pi^0$ /kaons/heavier mesons; muon  $p_\mu > 0.15$ , pion  $p_\pi > 0.175$  GeV/c, in the fiducial volume.

- Signal defined by the **final state**  $\Rightarrow$  every interaction mode contributes, via primary pion production *and* via FSI:

| Mode                                  | % of signal | Pion origin                         |
|---------------------------------------|-------------|-------------------------------------|
| RES ( $\Delta(1232)$ , higher $N^*$ ) | 77.5        | primary                             |
| DIS / SIS                             | 17.5        | primary (higher $W$ )               |
| QE                                    | 2.6         | <b>FSI only</b> (no primary $\pi$ ) |
| COH                                   | 2.1         | primary, low $ t $                  |
| MEC / 2p2h                            | 0.3         | <b>FSI only</b>                     |

- FSI also **migrates** events both ways: absorption/charge exchange turns multi- $\pi$  into  $1\pi$ , and  $1\pi$  into  $0\pi$ .
- NuMI off-axis:  $\nu_\mu$ -dominant FHC,  $\bar{\nu}_\mu$ -enriched RHC. Observables:  $p_\mu$ ,  $p_\pi$ ,  $\cos \theta_\mu$ ,  $\cos \theta_\pi$ ,  $\theta_{\mu\pi}$ .

## Exposure (fake data, blind)

|                    |                           |
|--------------------|---------------------------|
| FHC (Runs 1,2,4,5) | $8.86 \times 10^{20}$ POT |
| RHC (Runs 1,2,3,4) | $1.11 \times 10^{21}$ POT |
| Combined           | $1.99 \times 10^{21}$ POT |

# Event selection: cut sequence

A  $CC1\mu1\pi^\pm Xp$  topological selection: one muon, exactly one contained charged pion, any number of protons. Applied in order (cosmic rejection first, then PID, then kinematic cleanup):

1. **NeutrinoSlice** —  $\geq 1$  track, software trigger.
2. **InFiducialVol** — SCE-corrected  $\nu$  vertex in FV.
3. **Topological** — Pandora score  $> 0.67$ ; primary cosmic rejection (EXT  $\times 33$  down).
4. **MuonCandidate** —  $\geq 1$  primary track passes  $\mu$  PID; longest =  $\mu$ .
5. **ContainedPion** — exactly one other track passes  $\pi$  PID, endpoint contained; highest LLR =  $\pi$ .
- 6–7.  **$\mu/\pi$  In3Planes** — hit on each of U, V, Y.
8. **ShowerCut** — veto primary EM showers ( $\pi^0$  rejection; one soft shower tolerated).
9. **OpeningAngle** —  $\theta_{\mu\pi} < 2.6$  rad ( $149^\circ$ ); rejects large-angle cosmic/dirt.
10. **Nonprotons** (final) —  $< 4$  non-proton-like tracks (LLR  $> 0.1$ ) and  $< 5$  total primary tracks.

Per-run POT scaling; EXT (beam-off) supplies data-driven cosmics; dirt from the GENIE NuMI overlay. **Final: efficiency** 15.6%, **purity** 55% (measured phase space).

# Particle identification (muon & pion candidates)

Track-level PID combines Pandora track score, LLR calorimetric PID, Bragg-peak likelihoods and dedicated MIP/pion BDTs:

## Muon candidate (Cut 4)

| Variable        | Threshold     |
|-----------------|---------------|
| Track score     | $> 0.80$      |
| Vertex distance | $\leq 4.0$ cm |
| Track length    | $> 10.0$ cm   |
| LLR PID         | $> 0.20$      |
| MIP BDT         | $\geq -0.10$  |

## Pion candidate (Cut 5)

| Variable            | Threshold   |
|---------------------|-------------|
| LLR PID             | $> 0.10$    |
| Vertex distance     | $< 4.0$ cm  |
| Track length        | $> 20.0$ cm |
| MIP BDT             | $> -0.10$   |
| Pion BDT (vs. $p$ ) | $> -0.10$   |
| Bragg-pion score    | $\geq 0.08$ |
| Endpoint contained  | yes         |

Momentum reconstruction on the next slide.

# Momentum reconstruction: range, MCS and the proton range-KE

All three tracks are **range-based** where possible: a contained track's length maps onto momentum through the particle's  $dE/dx$ , which is far more precise than calorimetry at these energies.

| Particle          | Estimator                        | PeLEE branch         | Note                     |
|-------------------|----------------------------------|----------------------|--------------------------|
| $\mu$ (contained) | range momentum                   | trk_range_muon_mom_v | 37% of selected          |
| $\mu$ (exiting)   | MCS momentum                     | trk_mcs_muon_mom_v   | 63% — range invalid      |
| $\pi$             | range momentum, $\mu$ hypothesis | trk_range_muon_mom_v | $m_\mu$ close to $m_\pi$ |
| $p$               | range <b>KE</b> , $p$ hypothesis | trk_energy_proton_v  | → momentum, see below    |

$p_\mu$  is therefore a **hybrid** quantity (`..mom_reco` switches on muon containment) and is **MCS-dominated**: NuMI muons are energetic and the TPC is small, so most leave it.

## Proton: range KE → momentum

The proton branch gives a *kinetic energy*, so it is converted:

$$|\vec{p}_p| = \sqrt{T_p^2 + 2 T_p m_p}$$

Threshold  $|\vec{p}_p| > 0.3 \text{ GeV}/c$  ( $T_p \gtrsim 48 \text{ MeV}$ ) — below this the proton is too short to track. Performance: **−9% bias, 21% resolution**.

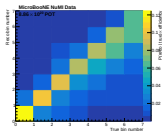
## Why the pion uses the $\mu$ hypothesis

$m_\mu = 105.7 \text{ MeV}$ ,  $m_\pi = 139.6 \text{ MeV}$ : the masses are close, so the range→momentum curve is a good proxy for a stopping pion.

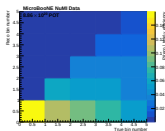
It also returns a **momentum**, directly comparable to the true  $p_\pi$  it is binned against — which keeps the migration matrix near-diagonal.

Valid only for **contained** tracks; the selection requires pion containment.

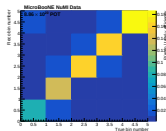
# Response (smearceptance) matrices — FHC



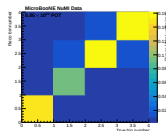
$p_\mu$



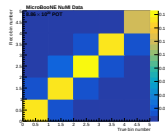
$p_\pi$



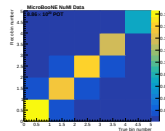
$\cos \theta_\mu$



$\cos \theta_\pi$



$\theta_{\mu\pi}$



$\theta_\mu$

$P(\text{reco}|\text{true}) \times \text{efficiency}$ ; columns are true bins, rows reco bins.  $p_\mu$  and the three angular observables are **near-diagonal** — note  $p_\mu$  is 63% MCS-reconstructed (exiting muons), whose poorer resolution sets its width.  $p_\pi$  is the **outlier** — reco piles into the first bin from every true bin, since pions interact hadronically before stopping so a range-based momentum **under-estimates**  $p_\pi$ . With 50–99% bin migration overall a plain matrix inversion is unusable, hence **Wiener-SVD** regularisation — at the cost of the  $A_C$  additional smearing.

**Planned scope change:** measured migration diagonals for  $p_\pi$  are 99/24/13/6/6% — the BNB CC1 $\pi$  criterion for a usable bin is  $> 68\%$ , so only the first passes. A golden-pion BDT was trained and reproduces the BNB classifier performance (43  $\rightarrow$  65% golden at 45% efficiency) but **cannot fix it**: even a **perfect** truth-level golden selection leaves bins 2–5 at 36/24/14/9%. We therefore expect to **withdraw**  $p_\pi$  from the **inclusive selection** and quote it on the golden-pion-enhanced and/or proton-tagged samples instead. All other observables are unaffected.

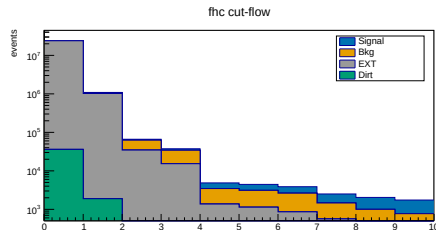
# Inclusive cut-flow — FHC (full exposure)

| Cut               | Signal     | Bkg        | EXT        | Eff%        | Pur%        |
|-------------------|------------|------------|------------|-------------|-------------|
| None              | 5 580      | 279 180    | 23.92M     | 100.0       | 0.02        |
| InFiducialVol     | 4 632      | 62 146     | 1.02M      | 83.0        | 0.42        |
| Topological       | 3 501      | 28 040     | 34 481     | 62.7        | 5.2         |
| MuonCandidate     | 3 143      | 18 980     | 15 206     | 56.3        | 8.3         |
| ContainedPion     | 1 395      | 2 094      | 1 348      | 25.0        | 28.5        |
| MuonIn3Planes     | 1 336      | 1 975      | 1 122      | 23.9        | 29.9        |
| PionIn3Planes     | 1 239      | 1 770      | 854        | 22.2        | 31.9        |
| ShowerCut         | 1 048      | 898        | 558        | 18.8        | 41.6        |
| OpeningAngle      | 1 040      | 813        | 186        | 18.6        | 50.9        |
| <b>Nonprotons</b> | <b>974</b> | <b>643</b> | <b>128</b> | <b>17.5</b> | <b>55.7</b> |

MC prediction, per-run POT scaled; data blind. Dirt  $\lesssim 1$  event at final cut.

## Final cut (Nonprotons), all beams:

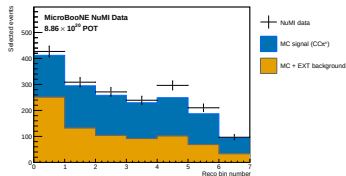
| Beam | Signal | Bkg   | EXT | Eff% | Pur% |
|------|--------|-------|-----|------|------|
| FHC  | 974    | 643   | 128 | 17.5 | 55.8 |
| RHC  | 1 081  | 746   | 133 | 17.6 | 55.1 |
| Comb | 2 055  | 1 389 | 261 | 17.5 | 55.4 |



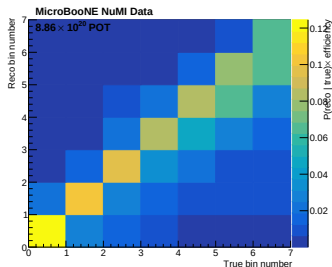
FHC stack (signal blue, bkg orange, EXT grey, dirt green; log scale). Two big drops: Topological (cosmics) and ContainedPion (single- $\pi$  topology).



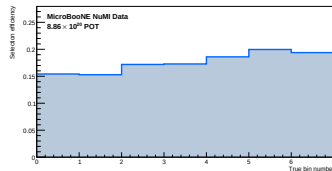
# Reconstruction, response & efficiency



Reco  $p_\mu$  (data/MC stack)



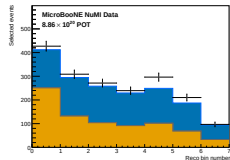
Response matrix



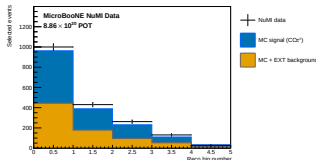
Efficiency vs  $p_\mu$

Wiener-SVD unfolding (framework WienerSVDUnfolder, 2nd-derivative regularisation) with the full systematic covariance from the universe throws.

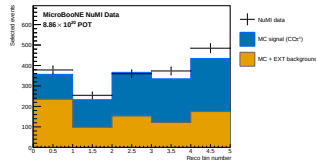
# FHC: selected events (POT-scaled), signal vs background



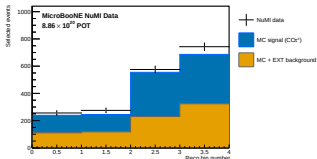
$p_\mu$



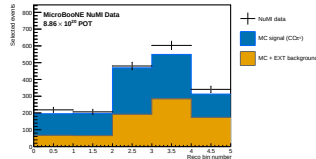
$p_\pi$



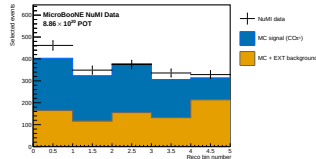
$\cos \theta_\mu$



$\cos \theta_\pi$



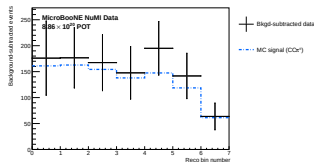
$\theta_{\mu\pi}$



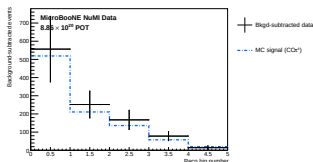
$\theta_\mu$

Flat reco-bin space, scaled to full FHC exposure. Points: (blind) fake data; stack: MC **signal** + **background**. Purity  $\simeq 55\%$  at the final cut.

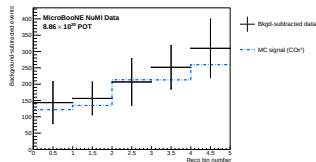
# FHC: after background subtraction



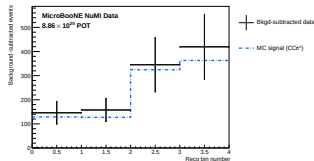
$p_\mu$



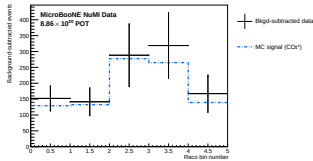
$p_\pi$



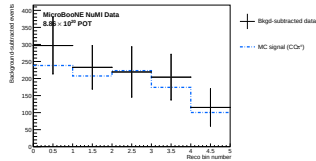
$\cos \theta_\mu$



$\cos \theta_\pi$



$\theta_\mu \pi$

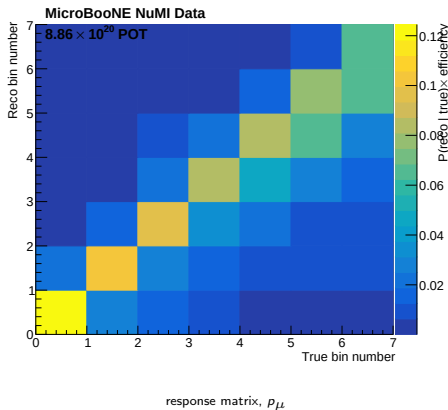


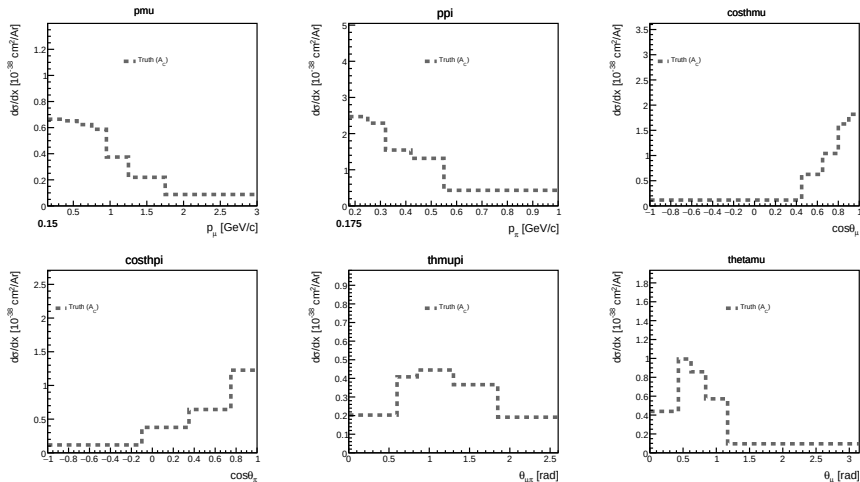
$\theta_\mu$

Data — estimated background vs CV MC signal — the input to the unfolding. Background from MC ( $\nu$ ) + beam-off EXT (cosmic) + dirt overlay; validated by the sidebands.

# FHC: unfolding

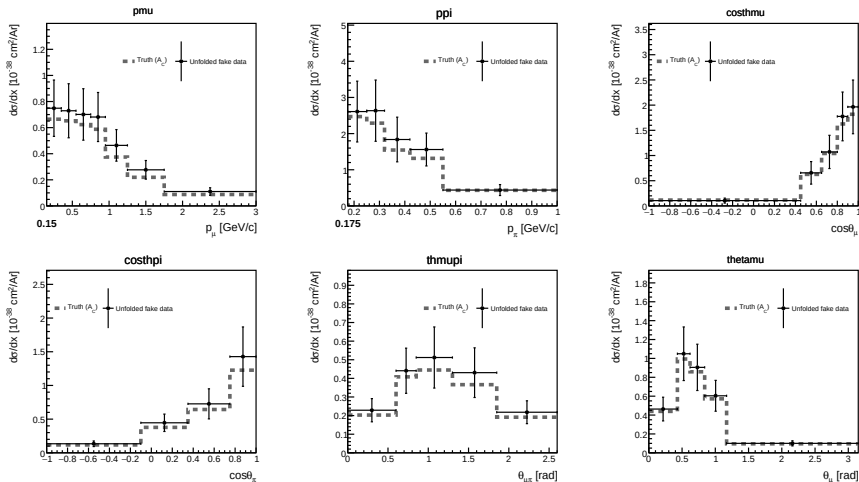
- ▶ The background-subtracted reco spectrum is **not** the cross section: it still carries the detector response and the selection efficiency.
- ▶ Unfolding inverts the response matrix  $A_{ij}$  shown earlier. With 50–99% bin migration a plain inversion is unusable, so the analysis uses **Wiener-SVD** regularisation.
- ▶ The price is the **additional-smearing matrix**  $A_C$ : the result is  $A_C \times (\text{true signal})$ , so **every** model must be multiplied by  $A_C$  before it is compared to the data.
- ▶ The next slides build up that comparison one curve at a time.





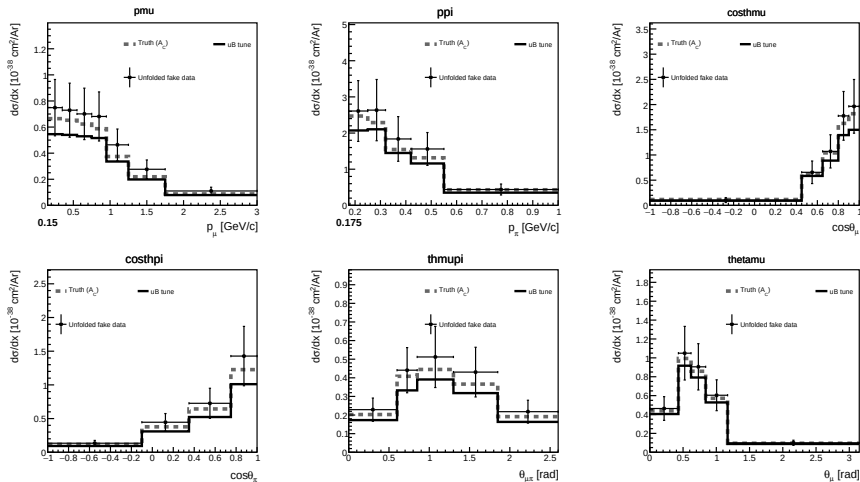
Poisson-thrown fake data are generated from the MicroBooNE tune; after unfolding we should recover this  $A_C$ -smeared truth.

# Differential cross sections — FHC: + unfolded fake data



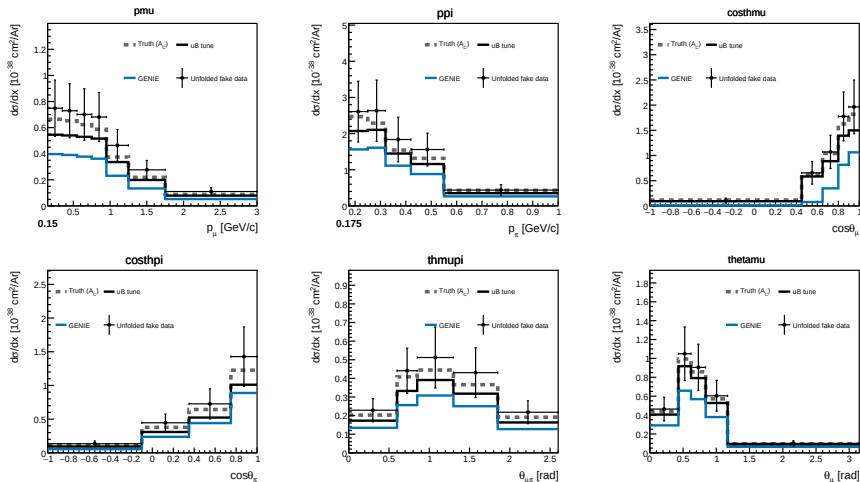
The unfolded points sit on the truth: the response, background subtraction and normalisation close.

# Differential cross sections — FHC: + MicroBooNE tune



The tune is the central-value model the fake data were thrown from.

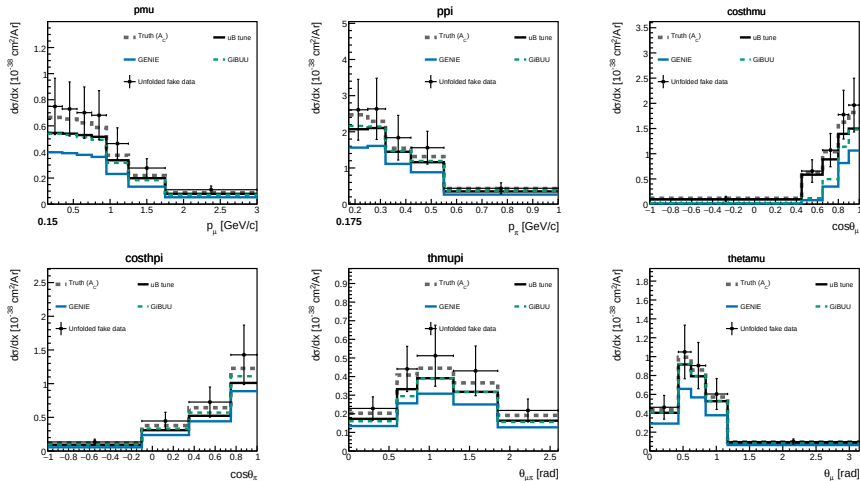
# Differential cross sections — FHC: + GENIE



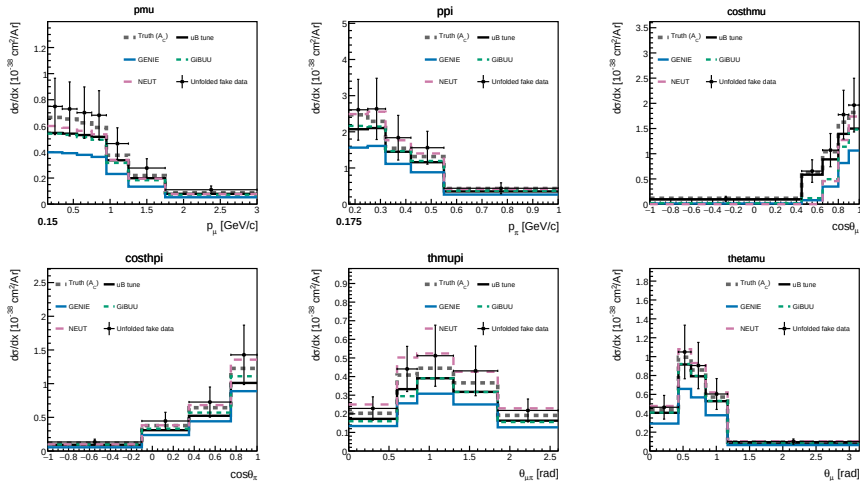
External generators enter, each  $A_C$ -smeared so it lives in the same space as the measurement.



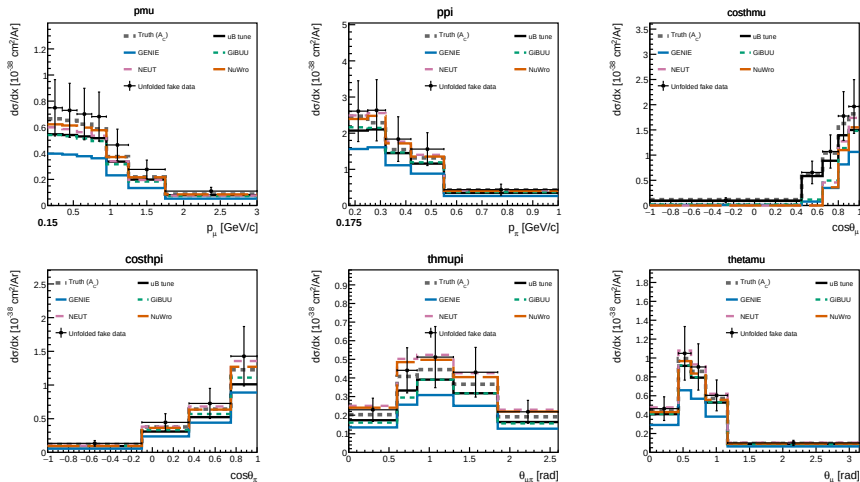
# Differential cross sections — FHC: + GiBUU



# Differential cross sections — FHC: $\pi^+$ NEUT

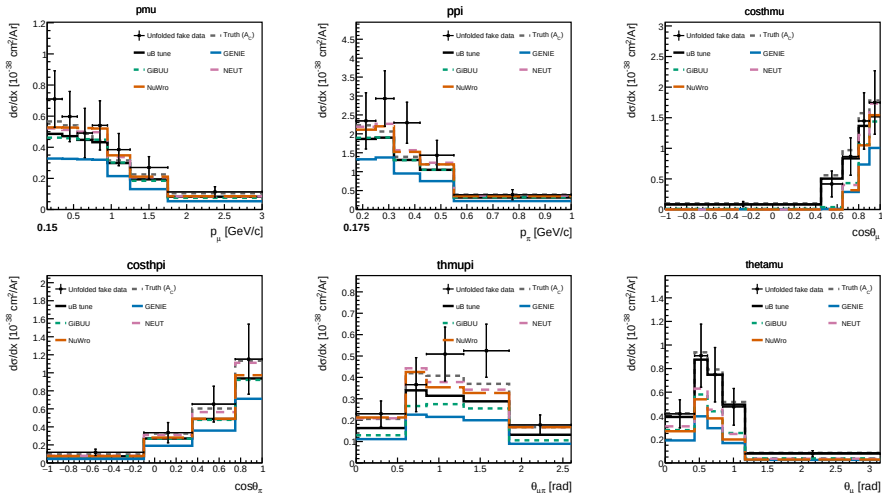


# Differential cross sections — FHC: + NuWro



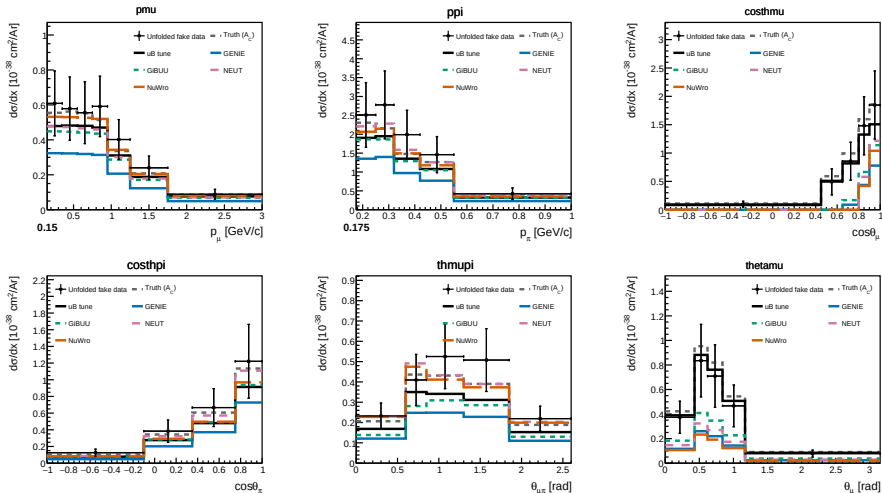
All four generators bracket the data. Error bars are stat  $\oplus$  syst: the full covariance (flux, cross-section, detector, re-interaction, POT, targets, data/MC/EXT stats) propagated through the unfolding —  $\simeq 27\%$  for  $p_\mu$  (FHC), flux-dominated (23%).

# Differential cross sections — RHC



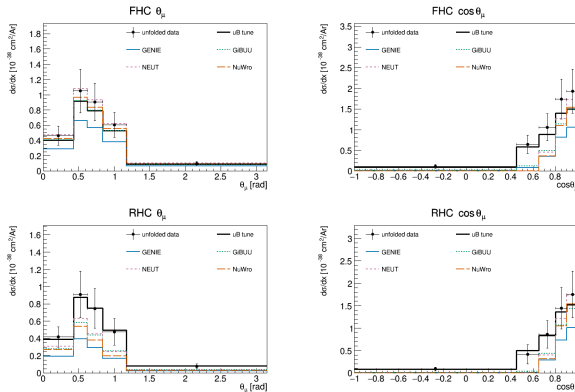
$\bar{\nu}_\mu$ -enriched beam; unfolded fake data vs  $A_C$ -smeared truth, MicroBooNE tune, and GENIE/GiBUU/NEUT/NuWro.

# Differential cross sections — combined



FHC+RHC combined exposure (per-universe correlated flux treatment). Error bars are  $\text{stat} \oplus \text{syst}$  (full covariance).

# Muon angle: $\cos \theta_\mu$ vs $\theta_\mu$



All curves  $A_C$ -smeared (same space as the data). Same events, same binning ( $\theta$  edges = arccos of the  $\cos \theta$  edges) — only the variable differs.

# $\cos \theta_\mu$ vs $\theta_\mu$ : the FHC $\theta_\mu$ “agreement” is an artefact

FHC  $\theta_\mu$ : normalised shapes

|         | bin   |       |       |       |       |
|---------|-------|-------|-------|-------|-------|
|         | 1     | 2     | 3     | 4     | 5     |
| data    | .1485 | .3361 | .2899 | .1934 | .0321 |
| truth   | .1484 | .3359 | .2901 | .1933 | .0323 |
| uB tune | .1484 | .3359 | .2901 | .1934 | .0322 |
| NEUT    | .1484 | .3359 | .2901 | .1934 | .0322 |

$A_C$  2nd/1st singular value

(same 4 generator inputs,  $s_2/s_1 = 5.0 \times 10^{-2}$ ):

|                       |                      |
|-----------------------|----------------------|
| FHC $\theta_\mu$      | $2.8 \times 10^{-6}$ |
| FHC $\cos \theta_\mu$ | $5.1 \times 10^{-2}$ |
| FHC $p_\mu$           | $6.6 \times 10^{-3}$ |
| RHC $\theta_\mu$      | $2.3 \times 10^{-2}$ |

- **Retraction.** We previously read the FHC  $\theta_\mu$   $\chi^2$  drop ( $\sim 26/5 \rightarrow \sim 2.8/5$ ) as evidence that  $\theta_\mu$  cures the  $\cos \theta_\mu$  tension. It does not.
- In FHC  $\theta_\mu$  the **data, the truth, the tune and all four generators share one shape to 4 decimals**.  $A_C$  there is **rank-1**: it annihilates every shape degree of freedom, so each model is mapped onto the same curve and only its normalisation survives.
- $\Rightarrow$  the agreement is **by construction**, effective ndf  $\approx 1$ , and  $\chi^2/5$  is not a goodness-of-fit. The same four generator inputs are **full rank** ( $s_2/s_1 = 5 \times 10^{-2}$ ), and  $\cos \theta_\mu/p_\mu$ /RHC  $\theta_\mu$  all retain their shape modes — so this is over-regularisation specific to FHC  $\theta_\mu$ , not a property of  $\theta$ .
- The earlier “ $A_C$  penalises the forward peak” explanation is also wrong: across observables  $A_C$  vs peakedness gives  $r = -0.17$  ( $W_{\pi p}$  sharpest yet 0.78;  $p_n$  flat yet **enhanced** 1.71), and within  $\cos \theta_\mu$  the **flattest** models (NEUT, NuWro) are suppressed **most**.
- **Action:** FHC  $\theta_\mu$  withdrawn from model comparison pending a regularisation re-run. All 18 extractions were re-unfolded with  $A_C$  now dumped to file — FHC  $\theta_\mu$  is the **only** rank-1 case (backup).

# Backup: how much model discrimination survives $A_C$ (all 18)

| Beam | Obs               | before | after  | keep |
|------|-------------------|--------|--------|------|
| FHC  | $p_\mu$           | .0248  | .0052  | 21%  |
| FHC  | $p_\pi$           | .0586  | .0080  | 14%  |
| FHC  | $\cos \theta_\mu$ | .0554  | .0724  | 131% |
| FHC  | $\cos \theta_\pi$ | .0589  | .0211  | 36%  |
| FHC  | $\theta_{\mu\pi}$ | .0422  | .0391  | 93%  |
| FHC  | $\theta_\mu$      | .0554  | 2.5e-6 | 0%   |
| RHC  | $p_\mu$           | .0232  | .0036  | 15%  |
| RHC  | $p_\pi$           | .0556  | .0068  | 12%  |
| RHC  | $\cos \theta_\mu$ | .0644  | .0771  | 120% |
| RHC  | $\cos \theta_\pi$ | .0629  | .0165  | 26%  |
| RHC  | $\theta_{\mu\pi}$ | .0505  | .0295  | 58%  |
| RHC  | $\theta_\mu$      | .0644  | .0188  | 29%  |
| COMB | $p_\mu$           | .0240  | .0028  | 12%  |
| COMB | $p_\pi$           | .0571  | .0064  | 11%  |
| COMB | $\cos \theta_\mu$ | .0598  | .5500  | 919% |
| COMB | $\cos \theta_\pi$ | .0609  | .0166  | 27%  |
| COMB | $\theta_{\mu\pi}$ | .0463  | .0360  | 78%  |
| COMB | $\theta_\mu$      | .0598  | .0070  | 12%  |

- ▶ “before/after” = largest pairwise difference between normalised generator shapes, truth-level vs  $A_C$ -smeared. Keep  $\approx 0 \Rightarrow$  the measurement can no longer tell the models apart.
- ▶ FHC  $\theta_\mu$  is the only zero — the only  $A_C$  whose rows are all one vector (spread 9%, eff. rank 1.02; everything else 170–1050%, 1.24–1.91). RHC/COMB  $\theta_\mu$  are **not** affected.
- ▶ Momenta keep only 11–21% everywhere: not destroyed, but a good  $\chi^2$  on  $p_\mu/p_\pi$  is a weak statement.  $\theta_{\mu\pi}$  (58–93%) is the healthiest variable we have.
- ▶ The  $> 100\%$  for  $\cos \theta_\mu$  is not good news:  $A_C$  pushes the generators **negative** in the backward bins while the tune stays positive (COMB bin 2: NuWro  $-0.43$  vs tune  $+0.51$ ). Unphysical  $\Rightarrow$  that spread is artefact, not discrimination, and its  $\chi^2$  inherits the problem.
- ▶ So the original framing was **backwards**: neither muon-angle variable is clean —  $\theta_\mu$  loses everything (FHC),  $\cos \theta_\mu$  keeps shape only via negative excursions.



# Integrated & total flux-averaged cross section

## Integrated $\sigma_{\text{int}}$ (Wiener-SVD)

|                   | FHC  | RHC  | Comb |
|-------------------|------|------|------|
| $p_\mu$           | 0.99 | 0.86 | 0.82 |
| $p_\pi$           | 0.96 | 0.97 | 0.96 |
| $\cos \theta_\mu$ | 0.82 | 0.66 | 0.70 |
| $\cos \theta_\pi$ | 0.97 | 0.81 | 0.86 |
| $\theta_{\mu\pi}$ | 0.88 | 0.88 | 0.92 |
| $\theta_\mu$      | 1.00 | 0.83 | 0.78 |
| $\theta_\pi$      | 0.97 | 0.80 | 0.86 |

spread = Wiener-SVD  $A_C$  smearing ( $10^{-38}\text{cm}^2/\text{Ar}$ )

## Total (cut-and-count, no unfolding)

| Config   | $\sigma$ ( $10^{-38}\text{cm}^2/\text{Ar}$ ) |
|----------|----------------------------------------------|
| FHC      | $1.06 \pm 0.27$                              |
| RHC      | $0.98 \pm 0.35$                              |
| Combined | $1.02 \pm 0.30$                              |

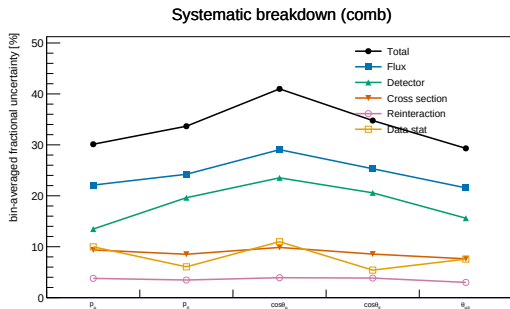
Uncertainties are **syst**  $\oplus$  **stat** (e.g. FHC: 25.5% systematic, 4.3% data-statistical). Cut-and-count carries **no**  $A_C$ , so it is directly comparable to the truth-level generator column.

## Generator totals: the spread is NOT $A_C$

|       | truth-level | $A_C$ -smeared | $A_C$ factor |
|-------|-------------|----------------|--------------|
| GENIE | 0.79        | 0.51           | 0.64         |
| GiBUU | 1.08        | 0.69           | 0.64         |
| NEUT  | 1.26        | 0.75           | 0.59         |
| NuWro | 1.20        | 0.81           | 0.67         |

$A_C$  is a **common**  $\simeq 0.6$  factor on all four models, so it cannot generate the  **$1.6\times$  spread** between them at truth level — that is

# Systematics & sideband validation



Flux (PPFX) dominates; detector second; total  $\sim 30$ – $40\%$ .

## Background-control sidebands (combined)

| Region          | $N_{\text{data}}$ | EXT    | sig% | data/ $\nu$ MC |
|-----------------|-------------------|--------|------|----------------|
| $\text{CC0}\pi$ | 24 148            | 17 855 | 9.7  | 1.00           |
| $\pi^0$         | 14 694            | 10 284 | 9.9  | 1.01           |
| multi- $\pi$    | 303               | 83     | 22.9 | 1.08           |
| cosmic          | 169               | 616    | 8.0  | 1.10           |

Each inverts **one** signal requirement.  $\text{CC0}\pi \rightarrow$  the FSI absorption path in/out of the signal;  $\pi^0 \rightarrow$  shower-veto efficiency; **multi- $\pi$**   $\rightarrow$  wrong-topology feed-down; **cosmic**  $\rightarrow$  the only EXT-dominated region ( $4.0\times$  the  $\nu$ MC).

Pseudo-data is a throw of the  $\nu$ MC, so data/ $\nu$ MC tests the **machinery** (per-run POT + weighting), not yet the background model. Small-region scatter is Poisson ( $\sqrt{N} \approx 8\%$  for cosmic).

At unblinding:  $\text{CC0}\pi/\pi^0$  constrain to  $\sim 1\%$ , well inside the 20–30% xsec systematic they test.

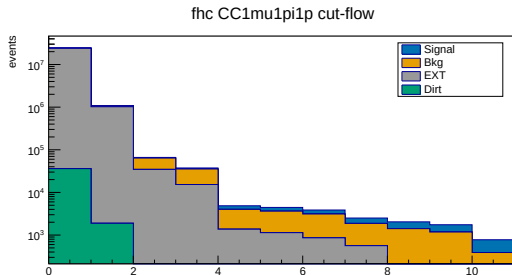
## Part II — Proton-tagged subsample: hadronic mass & TKI

### Idea

Requiring an identified proton reconstructs the full hadronic system, opening two new observable classes: the **hadronic invariant mass** (probes  $\Delta/N^*$  directly) and the **transverse kinematic imbalance** (Fermi motion + FSI).

- ▶ CC1 $\mu$ 1 $\pi$ 1 $p$ : inherits the inclusive selection + a leading proton (LLR PID,  $p_p \geq 0.3$  GeV/ $c$ , contained). Efficiency 11.5%, purity 48%.
- ▶ Six W/TKI observables:  $W_{\pi p}$ ,  $W_{\text{had}}$ ,  $\delta p_T$ ,  $\delta \alpha_T$ ,  $\delta \phi_T$ ,  $p_n$  (6 bins each) + five inclusive kinematic observables re-measured on the proton-tagged phase space.

# Selection performance & proton-tag quality



FHC proton-tagged cut-flow; last stage = proton identification

## Proton tag (truth-matched, FHC)

- ▶ 87% tag purity ( $\sim 13\%$  mis-tag)
- ▶ Rejects  $\sim 64\%$  background for  $\sim 30\%$  signal loss
- ▶ Purity  $33\% \rightarrow 48\%$ ;  $S/\sqrt{B} + 17\%$
- ▶ Proton  $p$ :  $-9\%$  bias,  $21\%$  resolution

# Proton-tagged cut-flow — FHC (full exposure)

Stages None–Nonprotons are the inherited inclusive selection (with the proton-tagged signal definition: inclusive signal + a true proton  $> 0.3$  GeV/c); the final **IdentProton** stage adds the leading-proton identification that defines the subsample.

| Cut                | Signal     | Bkg        | EXT       | Pred       | Eff%        | Pur%        |
|--------------------|------------|------------|-----------|------------|-------------|-------------|
| None               | 3 387      | 281 372    | 23.92M    | 24.7M      | 100.0       | 0.01        |
| InFiducialVol      | 2 834      | 63 944     | 1.02M     | 1.11M      | 83.7        | 0.26        |
| Topological        | 2 238      | 29 303     | 34 481    | 66 929     | 66.1        | 3.3         |
| MuonCandidate      | 2 031      | 20 092     | 15 206    | 37 776     | 60.0        | 5.4         |
| ContainedPion      | 852        | 2 637      | 1 348     | 4 890      | 25.2        | 17.4        |
| MuonIn3Planes      | 812        | 2 499      | 1 122     | 4 473      | 24.0        | 18.2        |
| PionIn3Planes      | 754        | 2 255      | 854       | 3 888      | 22.3        | 19.4        |
| ShowerCut          | 632        | 1 313      | 558       | 2 521      | 18.7        | 25.1        |
| OpeningAngle       | 628        | 1 226      | 186       | 2 045      | 18.5        | 30.7        |
| Nonprotons         | 568        | 1 049      | 128       | 1 749      | 16.8        | 32.5        |
| <b>IdentProton</b> | <b>401</b> | <b>373</b> | <b>46</b> | <b>828</b> | <b>11.8</b> | <b>48.4</b> |

IdentProton rejects  $\sim 64\%$  of the  $\nu$  background while keeping  $\sim 71\%$  of signal: purity  $32.5\% \rightarrow 48.4\%$ . RHC/comb: 46.6%/47.5% purity at  $\sim 12\%$  efficiency.

# The six proton-tagged observables

Built from the muon, charged-pion and leading-proton four-vectors ( $\vec{p}^{\text{had}} = \vec{p}_\pi + \vec{p}_p$ , transverse components w.r.t. the beam  $\hat{z}$ ; framework STV tool):

## Hadronic mass — probes $\Delta/N^*$

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2}$$
$$W_{\text{had}} = \sqrt{m_p^2 + 2m_p(E_{\text{cal}} - E_\mu) - Q^2}$$

$W_{\pi p}$  is **directly measured** — no assumption on the  $\nu$  direction or energy.  $W_{\text{had}}$  is calorimetric,  $E_{\text{cal}} = E_\mu + E_\pi + T_p$ .

Threshold:  $W_{\pi p} \geq m_\pi + m_p = 1.08 \text{ GeV}/c^2$ .

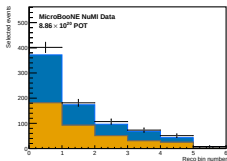
## TKI — probes Fermi motion & FSI

$$\delta p_T = |\vec{p}_T^\mu + \vec{p}_T^{\text{had}}|$$
$$\delta\alpha_T = \angle(-\vec{p}_T^\mu, \vec{p}_T^\mu + \vec{p}_T^{\text{had}})$$
$$\delta\phi_T = \angle(-\vec{p}_T^\mu, \vec{p}_T^{\text{had}})$$
$$p_n = \sqrt{\delta p_T^2 + p_L^2}$$

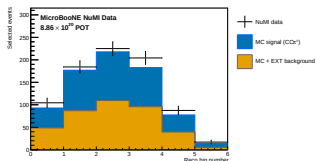
With no nuclear effects  $\delta p_T \rightarrow 0$ ; **Fermi motion broadens** it and **FSI drive  $\delta\alpha_T$  toward  $180^\circ$** .  $p_n$  infers the initial-state nucleon momentum ( $E_b = 24.78 \text{ MeV}$  argon removal energy enters  $p_L$ ).

Each is measured in six equal-population bins; the proton is required contained with  $|\vec{p}_p| > 0.3 \text{ GeV}/c$ .

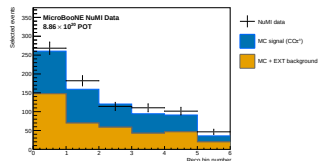
# Proton-tagged, FHC: selected events (POT-scaled)



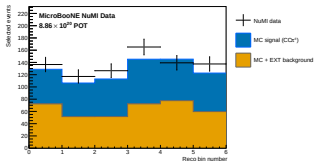
$W_{\pi p}$



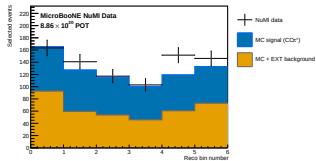
$W_{had}$



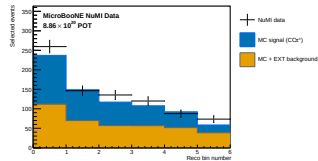
$\delta p_T$



$\delta \alpha_T$



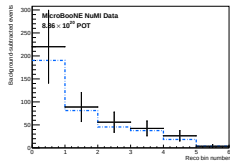
$\delta \phi_T$



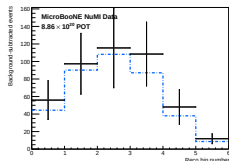
$\rho_n$

Points: (blind) fake data; stack: MC **signal** + **background**. Purity at the IdentProton stage is 48% — lower than the inclusive selection, so the subtraction is proportionally larger.

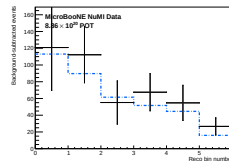
# Proton-tagged, FHC: after background subtraction



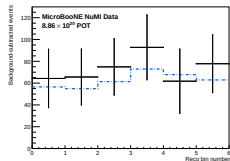
$W_{\pi p}$



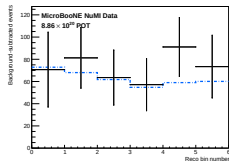
$W_{had}$



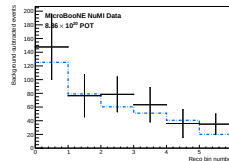
$\delta p_T$



$\delta \alpha_T$



$\delta \phi_T$

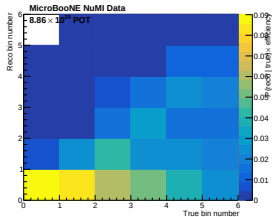


$\rho_n$

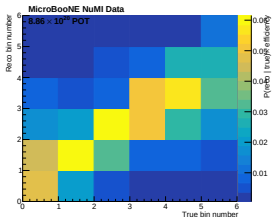
Data — estimated background vs CV MC signal — the input to the unfolding for the six W/TKI observables.



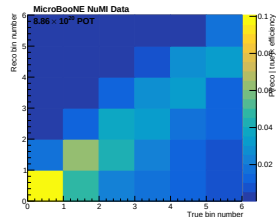
# Proton-tagged: response (smearceptance) matrices — FHC



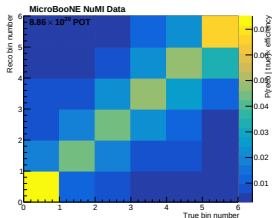
$W_{\pi p}$



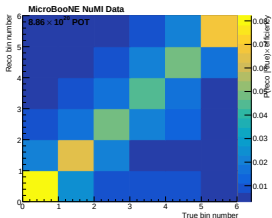
$W_{\text{had}}$



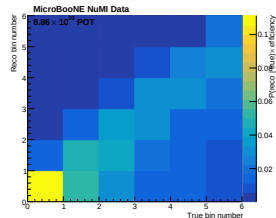
$\delta p_T$



$\delta \alpha_T$



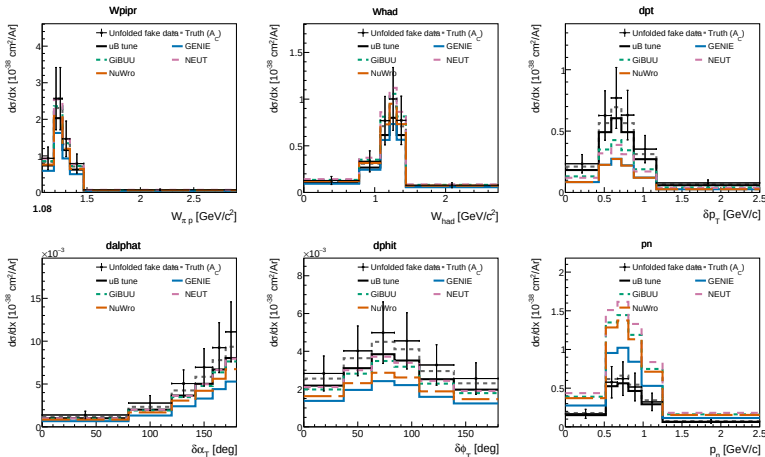
$\delta \phi_T$



$p_n$

$A_C$ :  $W_{\text{had}}$  0.88,  $W_{\pi p}$  0.78,  $\delta \alpha_T$  0.63,  $\delta \phi_T$  0.61,  $\delta p_T$  0.43,  $p_n$  1.71;  $\chi^2$  tracks  $|A_C - 1|$  (backup).

# Proton-tagged differential cross sections (FHC)



Unfolded fake data vs  $A_C$ -smeared truth, the uB tune and all four generators (predictions regenerated at the 6-bin analysis binning). Closure  $\chi^2/\text{ndf} \ll 1$  for every observable.

# Proton-tagged closure: all three beam configurations

| Observable        | $\sigma_{\text{int}} [10^{-38} \text{cm}^2/\text{Ar}]$ |       |       | COMB<br>$\chi^2/\text{ndf}$ |
|-------------------|--------------------------------------------------------|-------|-------|-----------------------------|
|                   | FHC                                                    | RHC   | COMB  |                             |
| $W_{\pi p}$       | 0.62                                                   | 0.578 | 0.564 | 0.07/6                      |
| $W_{\text{had}}$  | 0.60                                                   | 0.500 | 0.504 | 1.30/6                      |
| $\delta p_T$      | 0.61                                                   | 0.493 | 0.497 | 0.71/6                      |
| $\delta \alpha_T$ | 0.67                                                   | 0.567 | 0.667 | 0.86/6                      |
| $\delta \phi_T$   | 0.63                                                   | 0.568 | 0.573 | 0.05/6                      |
| $p_n$             | 0.52                                                   | 0.548 | 0.556 | 1.00/6                      |
| <hr/>             |                                                        |       |       |                             |
| $p_\mu$           | 0.56                                                   | 0.531 | 0.523 | 0.65/7                      |
| $p_\pi$           | 0.60                                                   | 0.572 | 0.582 | 2.49/5                      |
| $\cos \theta_\mu$ | 0.52                                                   | 0.512 | 0.529 | 0.52/5                      |
| $\cos \theta_\pi$ | 0.54                                                   | 0.444 | 0.454 | 2.17/4                      |
| $\theta_{\mu\pi}$ | 0.50                                                   | 0.410 | 0.447 | 0.21/5                      |

- ▶ All eleven observables now close positive in all three configurations — the proton-tagged measurement is complete. Every combined  $\chi^2$  is well below its number of bins.
- ▶ The RHC and combined results were previously negative and withdrawn. The cause was a POT-normalisation bug, not the data: the processed ntuples carry two POT conventions, and normalising by the sum of the distinct per-file exposures is correct for both.
- ▶ COMB lies between FHC and RHC for 7 of 11. The four exceptions are small ( $p_n$  +6.9% vs FHC;  $\cos \theta_\mu$  +1.7%;  $W_{\pi p}$  -2.4% and  $p_\mu$  -1.5% vs RHC) — COMB is an independent extraction with its own response, regularisation and  $A_C$ , not a fluence-weighted average, so it is not required to bracket. All four are far inside the  $\sim 30\%$  systematic.
- ▶ The 0.45–0.67 spread across observables is  $A_C$ , not physics: each observable's smearing rescales the integral differently, so these are not eleven independent measurements of one number.

# Generator comparison & background composition

## Integrated proton-tagged $\sigma$ ( $10^{-38}\text{cm}^2/\text{Ar}$ )

| Model                         | $\sigma$ (truth level) |
|-------------------------------|------------------------|
| GENIE                         | 0.50                   |
| NuWro                         | 0.65                   |
| GiBUU                         | 0.74                   |
| NEUT                          | 0.78                   |
| Unfolded data ( $W_{\pi p}$ ) | 0.62                   |

Spread  $\sim 1.6\times$ , largest in low- $\delta p_T$  / high- $p_n$  (FSI-sensitive).  
Generator values are truth-level (no  $A_C$ ); predictions  
regenerated at the 6-bin analysis binning, NuWro and NEUT  
from event vectors in the SL7 container.

## Remaining background (FHC)

| Component                           | %  |
|-------------------------------------|----|
| $\nu_\mu$ CC wrong topology         | 64 |
| True CC1 $\mu$ 1 $\pi$ , no $p>0.3$ | 18 |
| NC                                  | 11 |
| Out-of-FV                           | 5  |
| $\nu_e$ CC                          | 1  |

82% true  $\nu_\mu$  CC — well modelled & constrainable.

# Proton-tagged: background-control sidebands

The proton-tagged selection **inherits** the four sideband definitions from CC1 $\mu$ 1 $\pi$ Xp — no new regions or reprocessing needed. Each can be read either as inherited, or with a tagged proton additionally required (the measurement phase space).

| Region (comb) | $N_{\text{data}}$ | as inherited |      | $d/\nu\text{MC}$ |        | + proton tag |      | $d/\nu\text{MC}$ |
|---------------|-------------------|--------------|------|------------------|--------|--------------|------|------------------|
|               |                   | EXT          | sig% |                  |        | EXT          | sig% |                  |
| CC0 $\pi$     | 24 148            | 17 855       | 9.7  | 1.00             | 16 555 | <b>2 891</b> | 6.0  | 1.01             |
| $\pi^0$       | 14 694            | 10 284       | 9.9  | 1.01             | 9 019  | <b>2 370</b> | 7.1  | 1.02             |
| multi- $\pi$  | 303               | 83           | 22.9 | 1.08             | 137    | 36           | 18.9 | 1.07             |
| cosmic        | 169               | 616          | 8.0  | 1.10             | 60     | <b>12</b>    | 7.1  | 1.07             |

- ▶ **More** signal-depleted than for the inclusive selection (6–7% vs 48% purity in the signal region).
- ▶ The proton tag costs  $\sim 1/3$  of the statistics but cuts **EXT** by  $\sim 6\times$  — cosmits rarely have a reconstructed proton.
- ▶  $\Rightarrow$  the EXT normalisation systematic matters **much less** here than in the inclusive measurement.
- ▶ **Gap**: these *apply* the proton tag, so none constrains the  $\sim 13\%$  proton **mis-tag**. That needs an inverted-PID region (new branch + reprocess).

## Part III — Multi-pion: exclusive $\text{CC } 1\mu 2\pi$ and $\text{CC } 1\mu 3\pi$

New exclusive channels, building on the work of **Tusharadri Mahmud** (MSc) and **Linhui Gu**. Thin subselections of the inclusive selection: same muon, but *exactly* 2 or 3 charged pions (any number of protons). They probe the resonant/DIS transition and the intranuclear pion cascade, and are a background to the single-pion signal.

### Tuning vs. the $1\pi$ selection

| Setting              | $1\pi$      | Multi- $\pi$           |
|----------------------|-------------|------------------------|
| Pion-vertex distance | $\leq 4$ cm | $\leq 9.5$ cm (looser) |
| Uncontained pions    | 0           | up to 2                |
| Opening-angle cut    | on          | off                    |
| Shower veto          | on          | off                    |
| Wire-gap cuts        | on          | off                    |
| Per-pion threshold   | 0.175       | 0.10 GeV/c             |
| Pion ID              | LLR         | dedicated BDT          |

### Performance (Run 1 FHC only)

|                        | $2\pi$     | $3\pi$    |
|------------------------|------------|-----------|
| Efficiency             | 17.9%      | 10.1%     |
| Purity                 | 26.8%      | 11.3%     |
| Events @ full exposure | $\sim 670$ | $\sim 66$ |
| Reco ceiling           | 66.6%      | 52.4%     |

Efficiency/purity measured on the **Run-1 FHC  $\nu_\mu$  overlay alone** (one MC file); event yields are extrapolated to the full FHC exposure. **RHC is not yet processed** for the multi- $\pi$  channels.

The **reco ceiling** (fraction of true pions reconstructed as tracks at all) shows the loss is **particle-ID**, not reconstruction.

# Multi-pion: dedicated pion-ID BDT

- ▶ Inherited loose LLR cut replaced by a **dedicated BDT** (TMVA gradient boost), trained to separate true charged pions from the tracks polluting the multi- $\pi$  sample. **ROC = 0.81**.
- ▶ Inputs: pion/proton LLR, Bragg  $p/\mu$ /MIP likelihoods, track score, length, vertex distance — no explicit momentum.
- ▶  $2\pi$  uses **two momentum-specialised BDTs** (split at 20 cm track length, tighter soft-pion cut to protect the exact-multiplicity purity);  $3\pi$  keeps a **single BDT** (its pions are genuinely soft).
- ▶ Trialled on the **inclusive** selection it **hurts** (eff  $16.7 \rightarrow 18.9\%$  but purity  $60.3 \rightarrow 38.8\%$ )  $\Rightarrow$  inclusive keeps its LLR cut.

## Residual background (truth-matched)

| Counted “pion” really is       | %  |
|--------------------------------|----|
| real out-of-topology $\pi^\pm$ | 44 |
| proton mis-ID                  | 31 |
| muon                           | 12 |
| other                          | 13 |

Proton contamination is the **irreducible** core — a proton ranging out like a pion is genuinely ambiguous; adding a Bragg veto on top of the BDT did *not* reduce it and only cost efficiency.

Documented in a standalone technote; differential  $2\pi$  measurement + total  $\sigma$  for  $3\pi$  are the targets.

# Summary

- ▶ **Inclusive:** differential  $\text{CC1}\mu 1\pi^\pm$  cross sections for FHC, RHC and combined, per-run POT scaling, Wiener-SVD unfolding; total flux-averaged  $\sigma$  agrees with GENIE/GiBUU/NEUT/NuWro within  $\sim 1\sigma$ .
- ▶ **Proton-tagged:** new hadronic-mass ( $W_{\pi p}$ ,  $W_{\text{had}}$ ) and TKI ( $\delta p_T$ ,  $\delta\alpha_T$ ,  $\delta\phi_T$ ,  $p_n$ ) observables, plus the five inclusive kinematic observables re-measured. **Closure validated in all three configurations** at six-bin granularity — eleven observables  $\times$  FHC, RHC and combined, all positive, every combined  $\chi^2$  below its bin count.
- ▶ **Blind analysis:** fake data throughout; sidebands ( $\text{CC0}\pi$ , multi- $\pi$ ,  $\pi^0$ , cosmic) validated in all three beams, data/ $\nu$ MC  $\simeq 1.00$ , ready to constrain the background before unblinding.
- ▶  **$p_\pi$  scope change:** its response fails the standard binning criterion ( $>68\%$  diagonal) in 4 of 5 bins, and a trained golden-pion BDT — even at **perfect** truth-level selection — does not repair it. We expect to quote  $p_\pi$  on the **golden-pion-enhanced and/or proton-tagged** samples rather than the inclusive one. No other observable is affected.
- ▶ **Multi-pion:** an analysis of the exclusive  $2\pi$  and  $3\pi$  channels has started, building on the work of Tusharadri Mahmud (MSc) and Linhui Gu.

Thank you — questions?

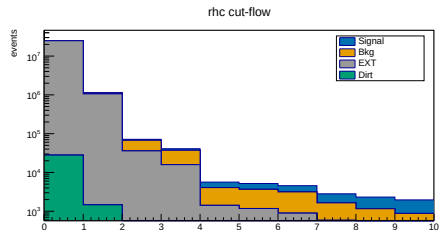


**Backup**

# Backup: inclusive cut-flow — RHC (full exposure)

| Cut               | Signal       | Bkg        | EXT        | Eff%        | Pur%        |
|-------------------|--------------|------------|------------|-------------|-------------|
| None              | 6 131        | 304 347    | 24.84M     | 100.0       | 0.02        |
| InFiducialVol     | 5 124        | 69 102     | 1.06M      | 83.6        | 0.44        |
| Topological       | 3 960        | 31 629     | 35 801     | 64.6        | 5.5         |
| MuonCandidate     | 3 559        | 21 443     | 15 788     | 58.0        | 8.6         |
| ContainedPion     | 1 577        | 2 649      | 1 400      | 25.7        | 27.8        |
| MuonIn3Planes     | 1 512        | 2 500      | 1 165      | 24.7        | 29.0        |
| PionIn3Planes     | 1 409        | 2 258      | 887        | 23.0        | 30.8        |
| ShowerCut         | 1 171        | 1 061      | 579        | 19.1        | 41.4        |
| OpeningAngle      | 1 162        | 965        | 193        | 19.0        | 50.0        |
| <b>Nonprotons</b> | <b>1 081</b> | <b>746</b> | <b>133</b> | <b>17.6</b> | <b>55.0</b> |

RHC (Runs 1,2,3,4a-c;  $1.108 \times 10^{21}$  POT). MC prediction, per-run POT scaled; data blind.

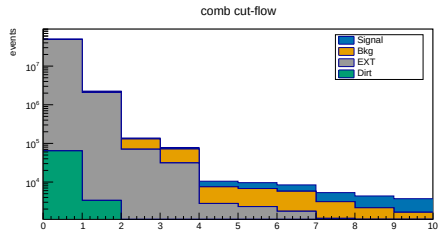


RHC stack (signal blue, bkg orange, EXT grey, dirt green; log scale).

# Backup: inclusive cut-flow — combined (full exposure)

| Cut               | Signal       | Bkg          | EXT        | Eff%        | Pur%        |
|-------------------|--------------|--------------|------------|-------------|-------------|
| None              | 11 710       | 583 527      | 48.76M     | 100.0       | 0.02        |
| InFiducialVol     | 9 756        | 131 249      | 2.08M      | 83.3        | 0.43        |
| Topological       | 7 461        | 59 669       | 70 282     | 63.7        | 5.4         |
| MuonCandidate     | 6 702        | 40 423       | 30 994     | 57.2        | 8.5         |
| ContainedPion     | 2 972        | 4 743        | 2 749      | 25.4        | 28.2        |
| MuonIn3Planes     | 2 847        | 4 475        | 2 287      | 24.3        | 29.4        |
| PionIn3Planes     | 2 648        | 4 028        | 1 742      | 22.6        | 31.3        |
| ShowerCut         | 2 218        | 1 959        | 1 137      | 18.9        | 41.5        |
| OpeningAngle      | 2 202        | 1 778        | 379        | 18.8        | 50.4        |
| <b>Nonprotons</b> | <b>2 055</b> | <b>1 389</b> | <b>261</b> | <b>17.5</b> | <b>55.4</b> |

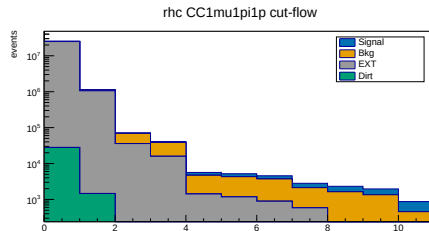
Combined ( $1.994 \times 10^{21}$  POT). Efficiency/purity track the two beams closely — the selection is beam-independent.



Combined stack (log scale).

# Backup: proton-tagged cut-flow — RHC (full exposure)

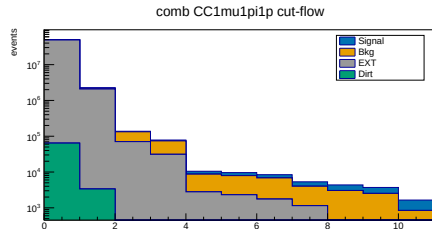
| Cut                | Signal     | Bkg        | EXT       | Eff%        | Pur%        |
|--------------------|------------|------------|-----------|-------------|-------------|
| None               | 3 638      | 306 840    | 24.84M    | 100.0       | 0.01        |
| InFiducialVol      | 3 064      | 71 162     | 1.06M     | 84.2        | 0.26        |
| Topological        | 2 455      | 33 134     | 35 801    | 67.5        | 3.4         |
| MuonCandidate      | 2 234      | 22 769     | 15 788    | 61.4        | 5.4         |
| ContainedPion      | 935        | 3 291      | 1 400     | 25.7        | 16.5        |
| MuonIn3Planes      | 892        | 3 119      | 1 165     | 24.5        | 17.1        |
| PionIn3Planes      | 831        | 2 835      | 887       | 22.9        | 18.2        |
| ShowerCut          | 687        | 1 544      | 579       | 18.9        | 24.3        |
| OpeningAngle       | 683        | 1 444      | 193       | 18.8        | 29.4        |
| Nonprotons         | 611        | 1 217      | 133       | 16.8        | 31.1        |
| <b>IdentProton</b> | <b>436</b> | <b>443</b> | <b>48</b> | <b>12.0</b> | <b>46.6</b> |



RHC proton-tagged stack; last stage IdentProton  
(leading-proton tag). Purity 31.1%  $\rightarrow$  46.6%.

# Backup: proton-tagged cut-flow — combined (full exposure)

| Cut                | Signal     | Bkg        | EXT       | Eff%        | Pur%        |
|--------------------|------------|------------|-----------|-------------|-------------|
| None               | 7 025      | 588 212    | 48.76M    | 100.0       | 0.01        |
| InFiducialVol      | 5 898      | 135 106    | 2.08M     | 84.0        | 0.26        |
| Topological        | 4 693      | 62 436     | 70 282    | 66.8        | 3.4         |
| MuonCandidate      | 4 265      | 42 860     | 30 994    | 60.7        | 5.4         |
| ContainedPion      | 1 787      | 5 928      | 2 749     | 25.4        | 16.9        |
| MuonIn3Planes      | 1 704      | 5 618      | 2 287     | 24.3        | 17.6        |
| PionIn3Planes      | 1 585      | 5 091      | 1 742     | 22.6        | 18.7        |
| ShowerCut          | 1 320      | 2 857      | 1 137     | 18.8        | 24.7        |
| OpeningAngle       | 1 310      | 2 670      | 379       | 18.7        | 30.0        |
| Nonprotons         | 1 179      | 2 266      | 261       | 16.8        | 31.8        |
| <b>IdentProton</b> | <b>837</b> | <b>816</b> | <b>95</b> | <b>11.9</b> | <b>47.5</b> |



Combined proton-tagged stack; IdentProton purity  
31.8%  $\rightarrow$  47.5%.

## Backup: per-run exposure

| Mode | Run | Data POT              | MC/data scale |
|------|-----|-----------------------|---------------|
| FHC  | 1   | $3.28 \times 10^{20}$ | 0.141         |
|      | 2   | $1.27 \times 10^{20}$ | 0.051         |
|      | 4   | $2.08 \times 10^{20}$ | 0.073         |
|      | 5   | $2.23 \times 10^{20}$ | 0.116         |
| RHC  | 1–4 | $1.11 \times 10^{21}$ | 0.045–0.091   |

Each run scaled by its own  $D_{\text{run}}/\text{MC}_{\text{run}}$  (real-data-ready).

## Backup: systematic breakdown (combined)

| Source        | $p_\mu$ | $p_\pi$ | $\cos \theta_\mu$ | $\cos \theta_\pi$ | $\theta_{\mu\pi}$ |
|---------------|---------|---------|-------------------|-------------------|-------------------|
| <b>Total</b>  | 30.1    | 33.7    | 41.0              | 34.8              | 29.3              |
| Flux (PPFX)   | 22.1    | 24.2    | 29.1              | 25.3              | 21.6              |
| Detector      | 13.5    | 19.6    | 23.5              | 20.6              | 15.6              |
| Cross section | 9.4     | 8.5     | 9.9               | 8.6               | 7.6               |
| Data stat     | 10.0    | 6.1     | 11.0              | 5.4               | 7.6               |

Bin-averaged fractional uncertainty [%]. Flux dominates throughout.

# Backup: history — the superseded pion-momentum estimator

An earlier version of this analysis reconstructed the **pion** momentum from the **proton-hypothesis range kinetic energy** (`trk_energy_proton_v`) rather than the muon-hypothesis range momentum used now.

- ▶ That is a **kinetic energy**, while the true axis it was binned against (with identical bin edges) is a **momentum** — two different physical quantities.
- ▶ Consequence: the migration matrix was **systematically shifted** rather than near-diagonal. A true 0.2 GeV/c pion gives  $KE \approx 0.104$  GeV, **two bins low**; a true 0.175 GeV/c pion falls **below the first reco bin entirely**.
- ▶ Fixed by switching to `trk_range_muon_mom_v` (`CC1mu1piXp.cxx`), which returns a momentum and exploits  $m_\mu \approx m_\pi$ .

Note: the proton-hypothesis range KE remains the *correct* estimator for the **proton** in the W/TKI subsample — there the quantity wanted really is a kinetic energy, converted to momentum via  $|\vec{p}_p| = \sqrt{T_p^2 + 2T_p m_p}$ .

Resolution panels produced before the switch quote a  $\sim 21\%$  underestimate of  $p_\pi$ ; they are to be regenerated with the current estimator.



# Backup: from PeLEE branches to the kinematic inputs

Every observable is built from three reconstructed tracks (muon, charged pion, leading proton), each taken from the PeLEE NeutrinoSelectionFilter ntuple at the candidate's PFP index:

| Quantity                              | PeLEE branch                                                                 | Note                                                              |
|---------------------------------------|------------------------------------------------------------------------------|-------------------------------------------------------------------|
| $ \vec{p}_\mu $                       | trk_range_muon_mom_v                                                         | range-based (primary)                                             |
| $ \vec{p}_\mu $ (x-check)             | trk_mcs_muon_mom_v                                                           | multiple-scattering                                               |
| $ \vec{p}_\pi $                       | trk_range_muon_mom_v                                                         | $\mu$ -hypothesis range, at the $\pi$ index                       |
| $T_p \rightarrow  \vec{p}_p $         | trk_energy_proton_v                                                          | $p$ -hypothesis range KE; $ \vec{p}_p  = \sqrt{T_p^2 + 2T_p m_p}$ |
| $\hat{u}_\mu, \hat{u}_\pi, \hat{u}_p$ | trk_dir_x/y/z_v                                                              | unit track direction                                              |
| $\cos \theta$                         | from $\hat{u}_z$                                                             | w.r.t. detector $z$ (beam-ish) axis                               |
| $\theta_{\mu\pi}$                     | $\arccos(\hat{u}_\mu \cdot \hat{u}_\pi)$                                     | opening angle                                                     |
| PID / candidacy                       | trk_llr_pid_score_v, trk_bragg*_v,<br>trk_score_v, trk_distance_v, trk_len_v | see PID slide                                                     |

Energies follow as  $E_i = \sqrt{|\vec{p}_i|^2 + m_i^2}$  with the PDG mass of the assigned species. Truth counterparts use mc\_nu\_daughter\_p[xyz] (back-matched via backtracked\_pdg).

## Backup: W/TKI observable definitions

From the muon, charged-pion and leading-proton four-vectors above (framework STV tool):

$$W_{\pi p} = \sqrt{(E_\pi + E_p)^2 - |\vec{p}_\pi + \vec{p}_p|^2} \quad (\text{direct, no } \nu \text{ direction})$$

$$W_{\text{had}} = \sqrt{m_p^2 + 2m_p(E_{\text{cal}} - E_\mu) - Q^2} \quad (\text{calorimetric})$$

$$\delta p_T = |\vec{p}_T^\mu + \vec{p}_T^{\text{had}}|, \quad \delta\alpha_T, \delta\phi_T \text{ (TKI angles)}, \quad p_n = \sqrt{\delta p_T^2 + p_L^2}$$

with  $\vec{p}^{\text{had}} = \vec{p}_\pi + \vec{p}_p$ ,  $E_{\text{cal}} = E_\mu + E_\pi + T_p$ , and the transverse components taken w.r.t. the beam ( $\hat{z}$ ) axis.  
 $E_b = 24.78$  MeV argon removal energy enters  $p_L$  (hence  $p_n$ ).

# Backup: proton-tagged binning & closure

| Observable       | Bin edges (6 equal-population bins)                 |
|------------------|-----------------------------------------------------|
| $W_{\pi p}$      | 1.08, 1.19, 1.23, 1.27, 1.34, 1.47, 2.00 GeV/ $c^2$ |
| $W_{\text{had}}$ | 0, 0.78, 1.08, 1.21, 1.31, 1.44, 1.90 GeV/ $c^2$    |
| $\delta p_T$     | 0, 0.43, 0.59, 0.72, 0.88, 1.17, 1.60 GeV/ $c$      |
| $\delta\alpha_T$ | 0, 80, 120, 142, 158, 170, 180 deg                  |
| $\delta\phi_T$   | 0, 37, 63, 84, 107, 139, 180 deg                    |
| $p_n$            | 0, 0.52, 0.67, 0.81, 0.98, 1.25, 1.70 GeV/ $c$      |

+ five kinematic observables ( $p_\mu$ ,  $p_\pi$ ,  $\cos\theta_\mu$ ,  $\cos\theta_\pi$ ,  $\theta_{\mu\pi}$ ) on inclusive binning.

FHC W/TKI closure  $\sigma_{\text{int}}$ : 0.62, 0.60, 0.61, 0.67, 0.63, 0.52;  $\chi^2/\text{ndf} \ll 1$ , all positive.

RHC closure also validated: all 11 observables positive,  $\sigma_{\text{int}}$  0.41–0.58, just below the FHC band as expected for the  $\bar{\nu}_\mu$ -enriched flux. Combined in production.

$A_C$  **smearing factor** (smeared/truth):  $W_{\text{had}}$  0.88,  $W_{\pi p}$  0.78 (sharpest distribution, yet barely affected — **no  $W$  reparameterisation needed**),  $\delta\alpha_T$  0.63,  $\delta\phi_T$  0.61,  $\delta p_T$  **0.43**,  $p_n$  **1.71 (enhanced)**. The last two distort model comparison in those panels.

# Backup: $\chi^2/\text{ndf}$ — inclusive, all observables and beams

Unfolded fake data compared with each  $A_C$ -smeared prediction.

| Beam | Observable        | truth  | uB tune | GENIE   | GiBUU   | NEUT    | NuWro   |
|------|-------------------|--------|---------|---------|---------|---------|---------|
| FHC  | $p_\mu$           | 1.66/7 | 3.09/7  | 5.99/7  | 3.87/7  | 3.77/7  | 2.90/7  |
| FHC  | $p_\pi$           | 1.96/5 | 3.77/5  | 5.05/5  | 3.78/5  | 3.33/5  | 3.46/5  |
| FHC  | $\cos \theta_\mu$ | 0.47/5 | 3.62/5  | 9.67/5  | 12.43/5 | 26.28/5 | 26.02/5 |
| FHC  | $\cos \theta_\pi$ | 0.49/4 | 1.20/4  | 3.82/4  | 0.99/4  | 1.34/4  | 1.19/4  |
| FHC  | $\theta_{\mu\pi}$ | 0.59/5 | 2.59/5  | 4.12/5  | 3.24/5  | 2.47/5  | 2.29/5  |
| RHC  | $p_\mu$           | 5.27/7 | 8.57/7  | 10.88/7 | 8.81/7  | 8.15/7  | 8.48/7  |
| RHC  | $p_\pi$           | 6.09/5 | 7.04/5  | 9.36/5  | 7.28/5  | 5.96/5  | 6.13/5  |
| RHC  | $\cos \theta_\mu$ | 0.96/5 | 4.91/5  | 7.45/5  | 8.66/5  | 18.43/5 | 17.42/5 |
| RHC  | $\cos \theta_\pi$ | 1.47/4 | 4.00/4  | 6.64/4  | 3.92/4  | 4.21/4  | 4.30/4  |
| RHC  | $\theta_{\mu\pi}$ | 4.32/5 | 5.57/5  | 7.67/5  | 5.68/5  | 6.56/5  | 7.59/5  |
| Comb | $p_\mu$           | 3.41/7 | 3.70/7  | 5.82/7  | 3.99/7  | 3.83/7  | 3.30/7  |
| Comb | $p_\pi$           | 1.32/5 | 2.32/5  | 3.85/5  | 2.65/5  | 1.82/5  | 2.05/5  |
| Comb | $\cos \theta_\mu$ | 2.49/5 | 4.61/5  | 9.07/5  | 12.02/5 | 26.72/5 | 27.20/5 |
| Comb | $\cos \theta_\pi$ | 0.42/4 | 1.83/4  | 4.01/4  | 1.69/4  | 1.89/4  | 1.95/4  |
| Comb | $\theta_{\mu\pi}$ | 2.10/5 | 2.42/5  | 3.91/5  | 3.05/5  | 3.49/5  | 3.45/5  |

The small *truth* column is the closure. RHC closure is visibly worse than FHC. The  $\cos \theta_\mu$  row is an outlier for NEUT/NuWro — see next slide.

# Backup: the $\cos \theta_\mu \chi^2$ is largely an $A_C$ artefact

Same events, same data, only the angular variable changes:

| Beam | Variable          | truth  | uB tune | GENIE  | GiBUU   | NEUT    | NuWro   |
|------|-------------------|--------|---------|--------|---------|---------|---------|
| FHC  | $\cos \theta_\mu$ | 0.47/5 | 3.62/5  | 9.67/5 | 12.43/5 | 26.28/5 | 26.02/5 |
| FHC  | $\theta_\mu$      | 0.05/5 | 3.07/5  | 4.73/5 | 3.03/5  | 2.82/5  | 2.90/5  |
| RHC  | $\cos \theta_\mu$ | 0.96/5 | 4.91/5  | 7.45/5 | 8.66/5  | 18.43/5 | 17.42/5 |
| RHC  | $\theta_\mu$      | 0.48/5 | 4.41/5  | 6.97/5 | 5.54/5  | 8.18/5  | 8.46/5  |
| FHC  | $\cos \theta_\pi$ | 0.49/4 | 1.20/4  | 3.82/4 | 0.99/4  | 1.34/4  | 1.19/4  |
| FHC  | $\theta_\pi$      | 0.44/4 | 1.15/4  | 3.77/4 | 0.93/4  | 1.30/4  | 1.18/4  |

In  $\cos \theta_\mu$  the  $A_C$  smearing suppresses the peaked generators by  $\times 0.33$  but the smoother tune by much less, so NEUT and NuWro are pushed far from the data and  $\chi^2$  explodes. In  $\theta_\mu$ , where the suppression is  $\times 0.76$ , the **same models agree well** ( $\chi^2$  falls by  $\sim 9\times$ ). The  $\pi$  angle, never sharply peaked, is unchanged. **The large  $\cos \theta_\mu \chi^2$  is therefore mostly a variable-choice artefact, not model disagreement** — a further argument for quoting  $\theta$  rather than  $\cos \theta$ .

## Backup: $\chi^2/\text{ndf}$ — proton-tagged (FHC)

| Observable       | truth  | uB tune | GENIE  | GiBUU   | NEUT    | NuWro   | $A_C$ |
|------------------|--------|---------|--------|---------|---------|---------|-------|
| $W_{\text{had}}$ | 0.09/6 | 1.39/6  | 1.65/6 | 1.06/6  | 1.17/6  | 1.05/6  | 0.88  |
| $W_{\pi p}$      | 0.09/6 | 0.95/6  | 1.78/6 | 0.62/6  | 0.56/6  | 0.87/6  | 0.78  |
| $\delta\alpha_T$ | 0.26/6 | 2.74/6  | 4.73/6 | 2.95/6  | 2.78/6  | 3.53/6  | 0.63  |
| $\delta\phi_T$   | 0.09/6 | 5.14/6  | 7.21/6 | 5.61/6  | 5.39/6  | 6.47/6  | 0.61  |
| $\delta p_T$     | 0.46/6 | 5.34/6  | 9.89/6 | 8.27/6  | 9.02/6  | 10.38/6 | 0.43  |
| $p_n$            | 0.09/6 | 4.24/6  | 7.72/6 | 18.74/6 | 25.40/6 | 16.45/6 | 1.71  |

Rows are ordered by how close the additional-smearing factor  $A_C$  is to unity, and the  $\chi^2$  follows it **monotonically**:  $|A_C - 1| = 0.12, 0.22, 0.37, 0.39, 0.57, 0.71$  against generator  $\chi^2 \simeq 1.2, 1.0, 3.3, 6.2, 9.4, 17$ . The two observables  $A_C$  barely touches ( $W_{\text{had}}, W_{\pi p}$ ) agree with **all four** generators; the two it distorts most ( $\delta p_T$  suppressed to 0.43,  $p_n$  *enhanced* to 1.71) show the largest  $\chi^2$  — in the same direction for every model. **This is the signature of a smearing artefact, not model discrimination**, the same effect seen in  $\cos\theta_\mu$ . The closure (truth column) is excellent throughout.

# Backup: generator predictions

- ▶ Inclusive: per-observable  $d\sigma/dx$  from GENIE (gst), GiBUU (FinalEvents), NEUT (neutvect), NuWro (event output), flux-combined to per-argon  $10^{-38} \text{ cm}^2$ .
- ▶ Proton-tagged W/TKI: same event samples reprocessed with the proton tag + TKI construction; NuWro and NEUT run inside the SL7 aptainer container.
- ▶ FTE rebinned to the coarsened analysis binning so every generator overlays on every observable.